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① Value functions

Q1] Given the expected return as:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad \cdots [1.1]$$

N.B: $r_t = R_t$

The the state-value function is defined as:

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s] \quad \cdots [1.2]$$

And the state-action value function is defined as

$$Q^{\pi}(s,a) = \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a] \cdots [1.3]$$

where Π is:

$$\pi: \mathcal{S} \rightarrow \mathcal{A}$$

$$\therefore \pi(a|s) = P(A_t = a | S_t = s) \quad \cdots [1.4]$$

Q2] [1.1] and [1.2] above define $V^{\pi}(s)$ and $Q^{\pi}(s,a)$ respectively.

By law of total expectation:

$$\mathbb{E}[X] = \sum_i \mathbb{E}[X|A_i] \cdot P(A_i)$$

$$\therefore V^{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s] = \sum_{a \in A} \left(\mathbb{E}_{\pi}[G_t | S_t = s, A_t = a] \cdot P(A_t = a | S_t = s) \right)$$

$$V^{\pi}(s) = \sum_{a \in A} Q^{\pi}(s,a)\, \pi(a|s) \quad \cdots [1.5]$$

For $Q^\pi(s,a)$ written in terms of $V^\pi(s)$

$$Q^\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

The expected return can be re-written as follows:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

$$= \gamma^0 R_{t+1} + \sum_{k=1}^{\infty} \gamma^k R_{t+k+1}$$

$$= R_{t+1} + \gamma \sum_{k=1}^{\infty} \gamma^{k-1} R_{t+k+1}$$

$$G_t = R_{t+1} + \gamma G_{t+1} \quad \cdots [1.6]$$

$$\therefore Q^\pi(s,a) = \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a]$$

Still applying law of total expectation over possible next-state and reward pairs:

$$Q^\pi(s,a) = \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a]$$

$$= \sum_{s' \in S} \sum_{r \in R} \left(\mathbb{E}_\pi \left[R_{t+1} + \gamma G_{t+1} \,\middle|\, \begin{matrix} S_t = s, A_t = a, \\ S_{t+1} = s', R_{t+1} = r \end{matrix} \right] \cdot p(s', r \mid s, a) \right)$$

$$= \sum_{\substack{(s', r) \\ \in \\ S \times R}} \left(\begin{matrix} \mathbb{E}_\pi \left[R_{t+1} \,\middle|\, \begin{matrix} S_t = s, A_t = a, \\ S_{t+1} = s', R_{t+1} = r \end{matrix} \right] \cdot p(s', r \mid s, a) \\ + \mathbb{E}_\pi \left[\gamma G_{t+1} \,\middle|\, \begin{matrix} S_t = s, A_t = a, \\ S_{t+1} = s', R_{t+1} = r \end{matrix} \right] \cdot p(s', r \mid s, a) \end{matrix} \right)$$

$\mathbb{E}_\pi[R_{t+1} \mid \cdots, R_{t+1} = r, \cdots] = r$

By Markov Property

$$\mathbb{E}_\pi[\gamma G_{t+1} \mid S_{t+1} = s', \cdots,] = \mathbb{E}_\pi[\gamma G_{t+1} \mid S_{t+1} = s']$$

$$\therefore Q^{\pi}(s,a) = \sum_{\substack{(s',r) \\ \in \\ S \times R}} \left(r + \gamma \mathbb{E}_{\pi}[G_{t+1} \mid S_{t+1} = s'] \right) \cdot p(s', r \mid s, a)$$

$$\therefore = \sum_{\substack{(s',r) \\ \in \\ S \times R}} [r + \gamma V^{\pi}(s')] \cdot p(s', r \mid s, a)$$

$$Q^{\pi}(s,a) = \sum_{s \in S} \sum_{r \in R} [r + \gamma V^{\pi}(s')] \cdot p(s', r \mid s, a) \quad \cdots [1.7]$$

→ To compute $Q^{\pi}(s,a)$ if $V^{\pi}(s')$ is known $\forall s' \in S$

- Multiply the state-value for every possible next state with the transition probability to that next state under the state-action pair (s,a)

$$\{ V^{\pi}(s') \cdot p(s' \mid s, a) \quad \forall s' \in S \}$$

- Then sum up the set of products above,

$$\sum_{s' \in S} V^{\pi}(s') \cdot p(s' \mid s, a)$$

and multiply by the discount factor γ

$$\gamma \sum_{s' \in S} V^{\pi}(s') \cdot p(s' \mid s, a)$$

- In add the immediate (expected) reward for the state-action pair (s,a) to the sum of products above to update $Q^{\pi}(s,a)$

$$Q^{\pi}(s,a) = R(s,a) + \gamma \sum_{s'} V^{\pi}(s') \cdot P(s'|s,a)$$

→ To compute $V^{\pi}(s)$ if the state-action value is known for all actions $a \in \mathcal{A}$ for that specific state $s \in S$

- for all possible actions $a \in \mathcal{A}$ in state $s \in S$, we compute the product of the probability of selecting action a in state s, $\pi(a|s)$, and the state-action value for the state-action pair $Q^{\pi}(s,a)$

$$\{ \pi(a|s) \cdot Q^{\pi}(s,a) \;\forall\; a \in \mathcal{A} \}$$

- We sum up the set of products above to update the state-value for the specified state $s \in S$

$$V^{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s)\, Q^{\pi}(s,a)$$

Q3]

(a) In a model free setting where the following are unknown:

- $R(s,a)$ → Reward function
- $P(s', r | s, a)$ → Transition probability
- $P(s'|s,a)$

It is more difficult to go from the state value, $V^{\pi}(s)$ to the state-action value, $Q^{\pi}(s,a)$.

In equation 1.7, assuming $V^{\pi}(s')$ is known for all $s' \in S$, the transition probability and immediate reward for the state-action pair remain undefined making it impossible

to compute $Q^{\pi}(s,a)$ in a model-free setting.

(b.)

For state s with two actions a_1 and a_2 such that:

$$\pi(a_1|s) = \pi(a_2|s) = 0.5$$

$$V^{\pi}(s) = 5$$

$$\therefore \quad V^{\pi}(s) = (0.5)Q^{\pi}(s,a_1) + (0.5)Q^{\pi}(s,a_2) \quad \cdots [1.8]$$

$$2V^{\pi}(s) = Q^{\pi}(s,a_1) + Q^{\pi}(s,a_2)$$

$$\therefore \quad Q^{\pi}(s,a_1) + Q^{\pi}(s,a_2) = 10$$

→ for pair-1 such that:

$$Q^{\pi}(s,a_1) > Q^{\pi}(s,a_2)$$

Choose:

$$Q^{\pi}(s,a_1) = 6$$

$$Q^{\pi}(s,a_2) = 4$$

Verification from equation [1.8]:

$$(0.5)6 + (0.5)4 = 3 + 2 = 5 = V_2^{\pi}(s)$$

→ for pair-2 such that:

$$Q^{\pi}(s,a_1) < Q^{\pi}(s,a_2)$$

Choose:

$$Q^{\pi}(s,a_1) = 3$$

$$Q^{\pi}(s,a_2) = 7$$

Verification from equation [1.8]:

$$(0.5)3 + 0.5(7) = 1.5 + 3.5 = 5 = V^{\pi}(s)$$

$\therefore$ $V^{\pi}(s)$ alone is insufficient to determine the value of

actions in the specific state as it maps all actions in that state to the same value, the state-value $V^{\pi}(s)$, comprising all individual action value into a single average value.

② Bellman Equations

ⓐ Bellman Expectation Equations

ⓐ Bellman Expectation Equation for state-value function, $V^{\pi}(s)$:

$$V^{\pi}(s) = \sum_{a \in A} \pi(a|s)\, Q^{\pi}(s,a)$$

$$V^{\pi}(s) = \sum_{a \in A} \pi(a|s) \left(\sum_{\substack{(s',r) \\ \in \\ S \times R}} p(s',r|s,a)\,[r + \gamma V^{\pi}(s')] \right)$$

... [2.1]

ⓑ Bellman Expectation Equation for state-action value function, $Q^{\pi}(s,a)$:

$$Q^{\pi}(s,a) = \sum_{\substack{(s',r) \\ \in \\ S \times R}} p(s',r|s,a)\,[r + \gamma V^{\pi}(s')]$$

$$Q^{\pi}(s,a) = \sum_{\substack{(s',r) \\ \in \\ S \times R}} p(s',r|s,a) \left[r + \gamma \sum_{a' \in A} \pi(a'|s')\, Q^{\pi}(s',a') \right]$$

... [2.2]

Q2] Contraction of Bellman Optimality Operator

r: $R^- \le r \le R^+$

γ: $0 \le \gamma \le 1$

$R(s,a) = R_{t+1} = r$

T^* : $Q(s,a) \rightarrow Q^*(s,a)$

$$: (T^*Q)(s,a) = R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) \cdot \max_{a' \in A} Q(s',a') \quad \cdots [2.3]$$

Equation [2.3] gives the Bellman optimality operator T^*.

$$\|Q\|_\infty = \max_{\substack{s \in S \\ a \in A}} |Q(s,a)| \quad \cdots [2.4]$$

To prove T^* is a γ-contraction under the norm:

$$\|T^*Q_1 - T^*Q_2\|_\infty \le \gamma \|Q_1 - Q_2\|_\infty \quad \cdots [2.5]$$

$$\rightarrow (T^*Q_1)(s,a) = R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) \max_{a' \in A} Q_1(s',a')$$

$$\rightarrow (T^*Q_2)(s,a) = R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) \max_{a' \in A} Q_2(s',a')$$

$$\therefore (T^*Q_1)(s,a) - (T^*Q_2)(s,a) = \gamma \sum_{s' \in S} \left(\max_{a' \in A} Q_1(s',a') - \max_{a' \in A} Q_2(s',a') \right) \cdot P(s'|s,a)$$

$$|(T^*Q_1)(s,a) - (T^*Q_2)(s,a)|$$

$$= \left| \gamma \sum_{s' \in S} P(s'|s,a) \left(\max_{a' \in A} Q_1(s',a') - \max_{a' \in A} Q_2(s',a') \right) \right|$$

$$\leq \gamma \left(\sum_{s' \in S} P(s'|s,a) \right) \left| \max_{a' \in A} Q_1(s',a') - \max_{a' \in A} Q_2(s',a') \right|$$

Using $\left| \max_a x_a - \max_a y_a \right| \leq \max_a |x_a - y_a| \quad \cdots [2.6]$

$$\gamma \sum_{s' \in S} P(s'|s,a) \left| \max_{a' \in A} Q_1(s',a') - \max_{a' \in A} Q_2(s',a') \right|$$

$$\leq \gamma \sum_{s' \in S} P(s'|s,a) \cdot \max_{a' \in A} \left| Q_1(s',a') - Q_2(s',a') \right|$$

$\therefore$

$$|(T^*Q_1)(s,a) - (T^*Q_2)(s,a)|$$

$$\leq \gamma \sum_{s' \in S} P(s'|s,a) \cdot \max_{a' \in A} \left| Q_1(s',a') - Q_2(s',a') \right|$$

$$\max_{a' \in A} |Q_1(s',a') - Q_2(s',a')| \leq \|Q_1 - Q_2\|_\infty$$

$\therefore$

$$|(T^*Q_1)(s,a) - (T^*Q_2)(s,a)| \leq \gamma \sum_{s' \in S} P(s'|s,a) \|Q_1 - Q_2\|_\infty$$

$$\leq \gamma \|Q_1 - Q_2\|_\infty \sum_{s' \in S} P(s'|s,a)$$

$$|(T^*Q_1)(s,a) - (T^*Q_2)(s,a)| \leq \gamma \|Q_1 - Q_2\|_\infty$$

$$\max_{s\in S,\, a\in A} \left| (T^*Q_1)(s,a) - (T^*Q_2)(s,a) \right| \le \max_{s\in S,\, a\in A} \gamma \|Q_1 - Q_2\|_\infty$$

$$\| T^*Q_1 - T^*Q_2 \|_\infty \le \gamma \|Q_1 - Q_2\|_\infty$$

Q3] Exact Policy Iteration

(a) Prove $\|Q^* - Q^{\pi_{k+1}}\|_\infty \le \|Q^* - Q^{\pi_k}\|_\infty$

(i) → By optimality Property

$$Q^* \ge Q^\pi \quad \forall \pi$$

$$\therefore Q^* - Q^\pi \ge 0 \quad \forall \pi$$

$$\therefore 0 \le Q^* - Q^{\pi_{k+1}} \quad \cdots [2.7]$$

→ By Policy Improvement Bound

$$T^{\pi_{k+1}} Q^{\pi_k} \le Q^{\pi_{k+1}}$$

$$\therefore Q^* - T^{\pi_{k+1}} Q^{\pi_k} \ge Q^* - Q^{\pi_{k+1}}$$

$$\therefore Q^* - Q^{\pi_{k+1}} \le Q^* - T^{\pi_{k+1}} Q^{\pi_k} \quad \cdots [2.8]$$

Combining Equations [2.7] and [2.8]

$$0 \le Q^* - Q^{\pi_{k+1}} \le Q^* - T^{\pi_{k+1}} Q^{\pi_k} \quad \cdots [2.9]$$

(ii) → By greedy Policy equation

$$T^{\pi_{k+1}} Q^{\pi_k} \ge T^* Q^{\pi_k}$$ ← greedy policy equation

$$T^* Q^* = Q^*$$ ← optimal fixed point equation

$$\therefore Q^* - T^{\pi_{k+1}} Q^{\pi_k} \geq Q^* - T^* Q^{\pi_k}$$
$$\geq T^* Q^* - T^* Q^{\pi_k}$$

$$\therefore \boxed{Q^* - T^{\pi_{k+1}} Q^{\pi_k} \geq T^* Q^* - T^* Q^{\pi_k}} \quad \cdots [2.10]$$

(ii) Combining Equations [2.9] and [2.10]

$$Q^* - Q^{\pi_{k+1}} \leq T^* Q^* - T^* Q^{\pi_k}$$

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \|T^* Q^* - T^* Q^{\pi_k}\|_\infty$$

from Equation [2.5]

$$\|T^* Q^* - T^* Q^{\pi_k}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty$$

$$\therefore \boxed{\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty} \quad \cdots [2.11]$$

The componentwise nonnegativity in step (i) (Equation [2.7]–[2.8]) permits norm comparison because the supremum norm defined in equation [2.4] takes the maximum on absolute values allowing us to compare for upper-bounds of supremum norm.

(b) Error after iteration k is defined as:

$$\ell_k = \|Q^* - Q^{\pi_k}\|_\infty$$

$$\therefore$$

$$\ell_0 = \|Q^* - Q^{\pi_0}\|_\infty$$
$$Q^{\pi_1} = TQ^{\pi_0}$$
$$Q^{\pi_n} = T^n Q^{\pi_0}$$

$\therefore$

$$e_k = \|Q^* - (T^*)^k Q^{\pi_0}\|_\infty$$

$$= \|(T^*)^k Q^* - (T^*)^k Q^{\pi_0}\|_\infty$$

$$\|(T^*)^k Q^* - (T^*)^k Q^{\pi_0}\|_\infty \leq \gamma^k \|Q^* - Q^{\pi_0}\|_\infty$$

$\therefore$

$$e_k \leq \gamma^k \|Q^* - Q^{\pi_0}\|_\infty$$

$$\boxed{e_k \leq \gamma^k e^0} \quad \text{— [2.12]}$$

Equation [2.12] also follows from equation [2.11]

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty$$

$\therefore$ $\|Q^* - Q^{\pi_k}\|_\infty \leq \gamma^k \|Q^* - Q^{\pi_0}\|_\infty$

$\therefore$ $\boxed{e_k \leq \gamma^k e^0}$ — [2.12]